Brighter Star Drilling Fluids

SAFETY DATA SHEET

BRSXTG — Xanthan Gum

Prepared in accordance with the Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

Section 1: Identification

Product identifier: BRSXTG

Chemical name / synonym: Xanthan gum

Recommended use: Viscosifier and suspending agent for freshwater- and brine-based drilling systems (slurry TBM, deep foundations, HDD)

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

Section 2: Hazards Identification

Emergency overview: Cream-colored, free-flowing powder with a flour-like odor

Warning statement: Caution! May cause skin, eye, and respiratory tract irritation

Acute eye: Slightly irritating. Dust may cause redness and irritation

Acute skin: Skin absorption not likely. May cause slight transient irritation

Acute inhalation: Dust may cause upper respiratory tract irritation

Acute ingestion: Non-toxic

Chronic effects: This product does not contain any ingredient designated by IARC, NTP, ACGIH, or OSHA as a probable or suspected human carcinogen

Section 3: Composition / Information on Ingredients

Chemical characterization: Substance.

Component	CAS No.
Xanthan gum	11138-66-2

Section 4: First-Aid Measures

Exposure Route	Action
Eye contact	Hold eyelids open and flush with a steady, gentle stream of water for at least 15 minutes. Seek medical attention if irritation develops or persists, or if visual changes occur
Skin contact	In case of contact, wash with plenty of soap and water. Seek medical attention if

Exposure Route	Action
	irritation develops or persists
Inhalation	Remove the victim from the immediate source of exposure and assure that the victim is breathing. If breathing is difficult, administer oxygen if available; if the victim is not breathing, administer CPR. Seek medical attention
Ingestion	Ingestion of the dry powder may cause the material to swell in the throat, possibly causing blockage and choking. If the victim is conscious and alert, give 1–2 glasses of water to prevent esophageal obstruction. Never give anything by mouth to an unconscious person. Seek medical attention. Do not leave the victim unattended

Conditions aggravated by exposure: Inhalation of the product may aggravate existing chronic respiratory problems such as asthma, emphysema, or bronchitis

Note to physician: All treatments should be based on the observed signs and symptoms of distress in the patient. Consideration should be given to the possibility that overexposure to materials other than this product may have occurred. Treat symptomatically; no specific antidote is available

Section 5: Fire-Fighting Measures

Flash point: > 93 °C (200 °F), closed cup. Flammability class: WILL BURN

Flammability limits: No data (lower / upper)

Extinguishing media: Small fires: carbon dioxide or dry chemical. Large fires: water or aqueous foam

Special procedures: Firefighters should wear NIOSH/MSHA-approved self-contained breathing apparatus and full protective clothing

Unusual fire and explosion hazards: The product will burn under fire conditions. Like all organic and most dry chemicals, this product (when mixed with air in critical proportions and in the presence of an ignition source) may present a dust explosion hazard

Hazardous decomposition products (under fire conditions): Oxides of carbon

Dust explosivity data:

Item	Value
Explosibility index	0.1 – 1 (type of explosion rated MODERATE)
Cloud ignition temperature	590 °C (1094 °F)
Minimum cloud ignition energy	> 10 mJ
Maximum explosion pressure	6.2 bar
Maximum rate of pressure rise	59 bar/s
Minimum explosion concentration	0.03 oz/ft³
Minimum ignition energy of dust cloud in air	> 999 mJ
Minimum autoignition temperature of dust clouds	400 °C (752 °F)

Item	Value
Hot-surface ignition temperature of dust layers	300 °C (572 °F)
Ignition sensitivity / explosion severity / layer ignition temperature	No data

Section 6: Accidental Release Measures

- CAUTION: Spilled material may become slippery when wet. Do not leave traces of product on floors, ladders, etc., as this may present a slipping hazard. Wear protective gear appropriate for the situation (see Section 8)
- Dry material: sweep up and place in an appropriate closed container. Wet material: absorb with an inert absorbent and shovel up into an appropriate closed container
- Spills should be reported to the competent local environmental authorities where required

Section 7: Handling and Storage

Handling: Avoid breathing dust. This product may present a dust explosion hazard: all dust-control equipment and material transport systems involved in handling this product should contain explosion relief vents, an explosion suppression system, or an oxygen-deficient environment. All conductive elements of the system that contact this material should be electrically bonded and grounded. The powder should not be flowed through non-conductive ducts or pipes. Use only appropriately classed electrical equipment

Storage: Keep the product in its original, tightly closed packaging in a cool, dry, and well-ventilated place; protect from moisture, rain, and direct sunlight. Inadequate storage may result in a loss of rheological properties

Section 8: Exposure Controls / Personal Protection

Exposure guidelines: No exposure limits have been established for this product or any of its ingredients

Engineering controls: Where engineering controls are indicated by use conditions or a potential for excessive exposure exists, traditional exposure control techniques may be used to effectively minimize employee exposure, such as wet processing methods to reduce dust generation

Respiratory protection: Select NIOSH/MSHA-approved equipment based on actual or potential airborne concentrations. Under normal conditions, an air-purifying (half-mask / full-face) respirator with cartridges or canister approved for use against dusts, mists, and fumes provides adequate protection

Eye / face protection: Select ANSI Z87-approved equipment appropriate for the particular use. As good practice, wear a minimum of safety glasses with side shields when working in industrial environments

Skin protection: Minimize skin contact through the use of gloves and suitable long-sleeved clothing (shirts and pants), giving consideration to both durability and permeation resistance

Work practice controls: Do not store, use, or consume foods, beverages, tobacco products, or cosmetics in areas where this material is stored. Wash hands and face carefully before eating, drinking, using tobacco, applying cosmetics, or using the toilet. Wash exposed skin promptly after accidental contact with this material

Section 9: Physical and Chemical Properties

Property	Value
Appearance	Cream-colored, free-flowing powder
Odor	Flour-like odor
pH (1% solution)	5.5 – 8.5
Water solubility	Soluble
Flash point (closed cup)	> 93 °C (200 °F)
Specific gravity	Not available
Melting point / boiling point	Not applicable
Vapor pressure / vapor density	Not applicable

Section 10: Stability and Reactivity

Chemical stability: This material is stable under normal handling and storage conditions as described in Section 7

Conditions to avoid: Dusting conditions, extreme heat, open flames, and sparks

Materials to avoid: Strong oxidizing agents

Hazardous decomposition products: Oxides of carbon (thermal decomposition)

Hazardous polymerization: Will not occur

Section 11: Toxicological Information

Acute toxicity (oral, dermal, inhalation): No test data found for the product; ingestion is considered non-toxic

Irritation (eye, skin, respiratory): No test data found for the product; dust may cause slight transient irritation (see Section 2)

Chronic toxicity / carcinogenicity: This product does not contain any substances considered by OSHA, NTP, IARC, or ACGIH to be "probable" or "suspected" human carcinogens. No additional test data found for the product

Section 12: Ecological Information

Ecotoxicological information: No data found for the product

Chemical fate information: No data found for the product

Section 13: Disposal Considerations

- Chemical additions, processing, or otherwise altering this material may make the waste management information presented in this SDS incomplete or inappropriate. Consult local regulations regarding the proper disposal of this material, as local requirements may be more restrictive or otherwise differ from national regulations
- Stabilize and solidify this material with compatible binders, then place in a secure landfill
- Any containers or equipment used should be decontaminated immediately after use

Section 14: Transport Information

TDG status: NON DANGEROUS

IMO status: NOT REGULATED

IATA status: NOT REGULATED

Note: The listed transportation classification does not address regulatory variations due to changes in package size, mode of shipment, or other regulatory descriptors.

Section 15: Regulatory Information

Inventory status (Y = all ingredients are on the inventory):

Inventory	Status
United States (TSCA)	Y
Canada (DSL)	Y
Europe (EINECS/ELINCS)	Y
Australia (AICS)	Y
Japan (MITI)	Y
South Korea (KECL)	Y

WHMIS classification: NOT CONTROLLED. This product has been classified in accordance with the hazard criteria of the CPR (Controlled Products Regulations), and this SDS contains all the information required by the CPR

Section 16: Other Information

NFPA ratings: Health — Slight; Flammability — Slight; Instability — Minimal

HMIS ratings: Health — Slight; Flammability — Slight; Reactivity — Minimal

Abbreviations: ACGIH — American Conference of Governmental Industrial Hygienists; OSHA — Occupational Safety and Health Administration; TLV — Threshold Limit Value; PEL — Permissible Exposure Limit; TWA — Time Weighted Average; STEL — Short Term Exposure Limit; NTP — National Toxicology Program; IARC — International Agency for Research on Cancer; ND — Not Determined

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